Algebraic Topology

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Chapter I

Homotopy

§1 Homotopy and homotopy equivalence

Definition 1.1. *Given a topological space X,*

1. *A **path** in X is a continuous map $\gamma : I \to X$ where $I = [0, 1]$ is the unit interval.*
2. *The points $\gamma(0)$ and $\gamma(1)$ are called the **initial point** and **terminal point** of the path, respectively. And $\gamma(0)$ and $\gamma(1)$ are also called the **endpoints** of the path.*
3. *A **loop** is a path γ such that $\gamma(0) = \gamma(1)$.*

Definition 1.2. *Two continuous maps $f, g : X \to Y$ are said to be **homotopic** if there exists a continuous map $H : X \times I \to Y$ such that $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$ for all $x \in X$. The map H is called a **homotopy** between f and g, and we write $f \simeq g$.*

Remark. *The relation of homotopy on paths with fixed endpoints in any space is an equivalence relation.*

Definition 1.3. *The set of all homotopy classes of loops at the base point x_0, denoted by*

$$\pi_1(X, x_0)$$

*is a group with respect to the product $[f][g] := [f * g]$. It is called the **fundamental group** of X at x_0.*

Proposition 1.4. *If there is a path h from x_0 to x_1, then there is an isomorphism $\pi_1(X, x_0) \to \pi_1(X, x_1)$ given by conjugation with $[f] \mapsto [h]^{-1}[f][h]$.*

Corollary 1.5. *If X is path-connected, then the fundamental group $\pi_1(X, x_0)$ is independent of the base point x_0 up to isomorphism.*

§1.1

Proposition 1.6. *The fundamental group* $\pi_1 : \mathbf{hTop}_* \to \mathbf{Grp}$ *is a covariant functor from the homotopy category of pointed topological spaces to the category of groups.*

Corollary 1.7. *If* X *retracts onto a subspace* A*, then* $i_* : \pi_1(A, x_0) \to \pi_1(X, x_0)$ *is injective.*

Chapter II

Homology

§1

Definition 1.1. *Given a sufficiently large N and consider the N-dimensional Euclidean space $\mathbb{R}^N$ and $v_0, v_1, \dots, v_n \in \mathbb{R}^N$.*

1. *A n-**simplex** is the convex hull of $n+1$ points $v_0, v_1, \dots, v_n$ in $\mathbb{R}^N$ that do not lie in any $(n-1)$-dimensional hyperplane. It is denoted by*

$$[v_0, v_1, \dots, v_n] := \left\{ \sum_{i=0}^{n} t_i v_i \mid t_i \geq 0, \sum t_i = 1 \right\}$$

   *v_i are called the **vertices** of the simplex.*

2. *A **subsimplex** of $[v_0, v_1, \dots, v_n]$ is a simplex of the form*

$$[v_{i_0}, v_{i_1}, \dots, v_{i_k}], \quad 0 \leq i_0 < i_1 < \cdots < i_k \leq n$$

   For any $k < n$ and k-subsimplex, there exists canonical linear homeomorphism

$$[v_0, v_1, \dots, v_k] \to [v_{i_0}, v_{i_1}, \dots, v_{i_k}], \quad t^j v_j \mapsto t^j v_{i_j}$$

3. *The subsimplex $[v_0, \dots, \hat{v}_i, \dots, v_n]$ is called the i-**th face** of $[v_0, v_1, \dots, v_n]$.*

Remark. *Without loss of generality, we can assume that $v_0 = 0$ and $v_i = e_i$ for $i = 1, \dots, n$, where e_i is the standard basis of $\mathbb{R}^n$. In this case, we denote the simplex by Δ^n and its i-th face by $\delta_i : \Delta^{n-1} \to \Delta^n$.*

Definition 1.2. *A Δ-complex structure on a space X is a collection of maps $\sigma_\alpha : \Delta^n \to X$ with n depending on α, such that*

(i) *Each restriction $\sigma_\alpha|_{\mathrm{Int}\,\Delta^n}$ is injective and $X = \bigsqcup_\alpha \sigma_\alpha(\mathrm{Int}\,\Delta^n)$*

(ii) *Each restriction of σ to a face of Δ^n is equal to some σ_β, that is, for each i, there exists β such that $\sigma_\alpha \circ d^i = \sigma_\beta$.*

(iii) *A set $A \subset X$ is open if and only if $\sigma_\alpha^{-1}(A)$ is open in Δ^n for each α.*

§2

Definition 2.1. *A CW complex is a space X together with a filtration*

$$\varnothing = X^{-1} \subset X^0 \subset X^1 \subset \cdots \subset X^n \subset \cdots \subset X$$

such that for each $n \geq 0$, there is a pushout diagram

$$\begin{tikzcd} \coprod_{\alpha\in\mathcal{A}_n} S_\alpha^{n-1} \arrow[r] \arrow[d, hook] & X^{n-1} \arrow[d] \\ \coprod_{\alpha\in\mathcal{A}_n} D_\alpha^n \arrow[r] & X^n \end{tikzcd}$$

A CW-complex is finite-dimensional if $X^n = X$ for some n; of finite type if each $\mathcal{A}_n$ is finite; and finite if it is finite-dimensional and of finite type.

§3

Definition 3.1. *A **singular n-simplex** in X is a continuous map $\sigma : \Delta^n \to X$. The set of all singular n-simplices in X is denoted by $\mathrm{Sin}_n(X)$. The **boundary operator** $\partial_n : \mathrm{Sin}_n(X) \to \mathrm{Sin}_{n-1}(X)$ is defined by*

$$\partial_n(\sigma) := \sum_{i=0}^{n} (-1)^i \sigma \circ \delta_i$$

Definition 3.2. *The abelian group of **singular n-chains** in X is the free abelian group $C_n(X) := \mathbb{Z}\mathrm{Sin}_n(X)$. The complex of singular chains is the sequence of abelian groups*

$$\cdots \xrightarrow{\partial_{n+1}} C_n(X) \xrightarrow{\partial_n} C_{n-1}(X) \xrightarrow{\partial_{n-1}} \cdots \xrightarrow{\partial_1} C_0(X) \xrightarrow{\partial_0} 0$$

The n-dimensional boundary $B_n(X) := \mathrm{Im}(\partial_{n+1})$ and the n-dimensional cycle $Z_n(X) := \ker(\partial_n)$ are subgroups of $C_n(X)$. The n-th singular homology group of X is defined as the quotient group $H_n(X) := Z_n(X)/B_n(X)$.

The singular simplex, singular chain, and singular homology are functorial. That is,

$$
\begin{tikzcd}
\textbf{Top} & & & \textbf{Ab} \\
X \arrow[d, "f"'] & \mathrm{Sin}(X) \arrow[d] & C_*(X) \arrow[d] & H_*(X) \arrow[d] \\
Y & \mathrm{Sin}(Y) & C_*(Y) & H_*(Y)
\end{tikzcd}
$$

where the map $f : X \to Y$ induces the maps.

$$
\begin{tikzcd}
\mathrm{Sin}_n(X) \arrow[r, "\partial_n"] \arrow[d, "f\circ -"'] & \mathrm{Sin}_{n-1}(X) \arrow[d, "f\circ -"] & \mapsto & C_n(X) \arrow[r, "\partial_n"] \arrow[d, "f_\#"'] & C_{n-1}(X) \arrow[d, "f_\#"] & \mapsto & H_n(X) \arrow[r, "\partial_n"] \arrow[d, "f_*"'] & H_{n-1}(X) \arrow[d, "f_*"] \\
\mathrm{Sin}_n(Y) \arrow[r, "\partial_n"] & \mathrm{Sin}_{n-1}(Y) & & C_n(Y) \arrow[r, "\partial_n"] & C_{n-1}(Y) & & H_n(Y) \arrow[r, "\partial_n"] & H_{n-1}(Y)
\end{tikzcd}
$$

In particular, Sin_n, C_n and H_n are also functors from **Top** to **Ab** or **Set**. And the natural transformations between H_n and H_{n-1} is the boundary operators ∂_n.

Theorem 3.3. *If two maps $f, g : X \to Y$ are homotopic, then they induce the homotopic chain maps, and same homomorphism.*

Proposition 3.4. *If decomposition $X = \coprod_\alpha X_\alpha$ is a disjoint union of open subsets(especially its path components), then $H_n(X) \cong \prod_\alpha H_n(X_\alpha)$ for each $n \geq 0$.*

Proposition 3.5. *If X is nonempty and path-connected, then $H_0(X) \cong \mathbb{Z}$. Hence for any space X, $H_0(X)$ is a free abelian group.*

§4 Axioms of Singular Homology Theory

Let X be a topological space and $A \subset X$ be a subspace. The pair (X, A) is called a **topological space pair**. A map of space pairs $f : (X, A) \to (Y, B)$ is a continuous map $f : X \to Y$ such that $f(A) \subset B$. A continuous map $f : X \to Y$ can be viewed as a map of pairs $f : (X, \varnothing) \to (Y, \varnothing)$. This is the category of space pair $\mathbf{Top}^2$ whose objects are topological pairs and morphisms are maps of space pairs. The maps $f, g : (X, A) \to (Y, B)$ are said to be **homotopic** if there is a homotopy $H : X \times I \to Y$ such that $H(A \times I) \subset B$ and $H(x, 0) = f(x)$, $H(x, 1) = g(x)$ for all $x \in X$.

For every n, there is a abelian group

$$H_n(X, A)$$

defined as $H_n(X)/H_n(A)$ (The notation $H_n(X)$ denotes $H_n(X, \varnothing)$). The continuous map f :

$(X, A) \to (Y, B)$ induces a homomorphism

$$f_* : H_n(X, A) \to H_n(Y, B)$$

for each n and satisfies the functoriality. In oter words, we define functor

$$H_n : \mathbf{Top}^2 \to \mathbf{Ab}$$

for each n.

For each n, there is a natural transformation ∂ such that the following diagram commutes

$$\begin{tikzcd} H_n(X,A) \arrow[r, "\partial"] \arrow[d, "f_*"'] & H_{n-1}(A) \arrow[d, "f_*"] \\ H_n(Y,B) \arrow[r, "\partial"] & H_{n-1}(B) \end{tikzcd}$$

Theorem 4.1 (Long exact sequence)**.** *Given a space pair (X, A), there is a exact sequence of chain complexes $0 \to C_*(A) \xrightarrow{i} C_*(X) \xrightarrow{j} C_*(X, A) \to 0$ where $C_n(X, A) := C_n(X)/C_n(A)$. Thus there is a long exact sequence of homology groups*

$$\cdots \to H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \to \cdots$$

Proposition 4.2. *If $f, g : (X, A) \to (Y, B)$ are homotopic maps, then they induce the same homomorphism on homology: $f_* = g_*$.*

Theorem 4.3 (Excision)**.** *Given subspaces $Z \subset A \subset X$ such that $\overline{Z} \subset \mathrm{Int}(A)$, then the inclusion $(X - Z, A - Z) \hookrightarrow (X, A)$ induces an isomorphism $H_n(X - Z, A - Z) \cong H_n(X, A)$ for all n.*

§5 The universal coefficient theorem

Theorem 5.1.

$$0 \longrightarrow H_n(X) \otimes G \longrightarrow H_n(X; G) \longrightarrow \mathrm{Tor}(H_{n-1}(X), G) \longrightarrow 0$$

§6 Mayer-Vietoris sequence

Theorem 6.1. *If $X = A \cup B$ where the interiors of A and B cover X, then there is a long exact sequence*

$$\cdots \longrightarrow H_n(A \cap B) \longrightarrow H_n(A) \oplus H_n(B) \longrightarrow H_n(X) \longrightarrow H_{n-1}(A \cap B) \longrightarrow \cdots$$

§7 Kunneth formula

Theorem 7.1 (Eilenberg-Zilber)**.**

$$C_*(X \times Y) \cong C_*(X) \otimes C_*(Y)$$

is a chain homotopy equivalence.

Theorem 7.2 (Kunneth formula)**.**

$$0 \longrightarrow \bigoplus_{p+q=n} H_p(X) \otimes H_q(Y) \longrightarrow H_n(X \times Y) \longrightarrow \bigoplus_{p+q=n-1} \mathrm{Tor}(H_p(X), H_q(Y)) \longrightarrow 0$$

Chapter III

Cohomology

§1 Singular cohomology

Definition 1.1. *Consider the R-coefficient chain complex $C_n(X;R)$, the R-coefficient cochain complex is defined as $C^n(X;R) := \mathrm{Hom}_R(C_n(X;R), R)$*

$$C_{n+1} \xrightarrow{\partial} C_n \xrightarrow{\partial} C_{n-1}$$

$$C^{n+1} \xleftarrow{\delta} C^n \xleftarrow{\delta} C^{n-1}$$

The singular cohomology group is defined as

$$H^n(X;R) := \mathrm{Ker}\,()$$

Theorem 1.2.

§1.1 Cup product and cohomology ring

Definition 1.3. *For cochains $\phi \in C^p(X;R)$ and $\psi \in C^q(X;R)$, the **cup product** $\phi \smile \psi$ is defined as*

$$\phi \smile \psi(\sigma) := \phi(\sigma|_{[v_0,\ldots,v_p]}) \cdot \psi(\sigma|_{[v_p,\ldots,v_{p+q}]})$$

Lemma 1.4. *For cochains $\phi \in C^p(X;R)$,$\psi \in C^q(X;R)$ and $\varphi \in C^r(X;R)$, the following properties hold:*

1. *Bilinearity*
2. *Associativity: $(\phi \smile \psi) \smile \varphi = \phi \smile (\psi \smile \varphi)$*
3. *The identity element: $1 \in C^0(X;R)$ is the identity element for the cup product.*

Definition 1.5. *The **cohomology ring** $H^*(X;R)$ is the graded ring defined as*

$$H^*(X;R) := \bigoplus_{n=0}^{\infty} H^n(X;R)$$

with the cup product as the multiplication.

Remark. *It has a natural R-module structure.*

Definition 1.6. *The **cross product** or **external cup product***

$$\times : H^p(X;R) \times H^q(Y;R) \longrightarrow H^{p+q}(X \times Y;R)$$

is defined as

$$\phi \times \psi := p_1^*\phi \smile p_2^*\psi$$

where $p_1 : X \times Y \to X$ and $p_2 : X \times Y \to Y$ are the projection maps.

Proposition 1.7. *1. $\delta(\phi \smile \psi) = \delta\phi \smile \psi + (-1)^p \phi \smile \delta\psi$*

2. $\delta(\phi \smile \psi) = (-1)^{p+q}\delta(\psi \smile \phi)$

3. δ has naturity: $f^*(\phi \smile \psi) = f^*\phi \smile f^*\psi$